

Project Workflow: Week 1

Phase: Project Planning & Concept Finalization

1. Objective

The primary goal of Week 1 was to establish a robust foundation for a classic, interactive Ping Pong game delivering a smooth and engaging user experience. This included:

- Defining the project scope and core gameplay boundaries.
- Selecting an optimal Java-based technology stack for responsive real-time rendering and game logic.
- Conceptualizing a clean, minimalist UI that keeps the player's focus on the game.

2. Detailed Task Breakdown

Feature Identification

Conducted research on classic arcade Pong implementations to finalize the core feature set: smooth paddle and ball movement, responsive two-player controls, real-time score tracking, and a clean modern visual style.

Technical Stack Selection

Selected Java with Object-Oriented Programming principles for clean state management, Swing for GUI rendering and the game canvas, the Timer class for the frame-rate-controlled game loop, and KeyListener for responsive, real-time paddle controls.

UI/UX Design

Created wireframes for the game canvas, paddle layout, and score display, establishing a minimalist, modern aesthetic that keeps the visual focus entirely on the gameplay.

Architectural Setup

Designed a modular, object-oriented class architecture separating the game loop, ball physics, paddle/collision logic, and score tracking to ensure clean code separation between rendering and game logic.

3. Deliverables

- Project Scope Document: A formal outline of the game's features and limitations.
- UI Wireframe Sketches: Conceptual layouts for the canvas, paddles, and score display.
- Tech Stack Definition: Finalized choice of Java, Swing, and the Timer-based game loop.
- Project Timeline: A structured 10-week development roadmap.

4. Tools & Resources

Development Environment

- Java JDK 11+
- IntelliJ IDEA / VS Code

Reference Materials

Java Swing, AWT, and Timer-based game loop references.